

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2022

TEST DATE: 06-02-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A
Sol. Direction of propagation is $\vec{E} \times \vec{B}$
2. B
Sol. Peak value of information voltage = 18 V
$$18 = \frac{\text{modulation percentage} \times \text{peak carrying voltage}}{100}$$
$$V_C = 30 \text{ V}$$
$$\text{Peak carrier voltage } V_C = \frac{V_{\max} + V_{\min}}{2}$$
$$\therefore V_{\max} + V_{\min} = 60 \text{ V}$$
3. C
Sol. Lyman series $\rightarrow$ UV region
Highest $\lambda \Rightarrow n = 2 \rightarrow n = 1$
 $E_2 - E_1 = 10.2 \text{ eV}$
 $k_{\max} = 10.2 \text{ eV} - \phi$
 $0 = 10.2 \text{ eV} - \phi$ (just able to eject)
$$v_{\text{th}} = \frac{\phi}{h} = \frac{10.2 \text{ eV}}{h} = 2.46 \times 10^{15} \text{ Hz}$$
4. C
Sol. $v_x = u \cos \theta$, $v_y = u \sin \theta - gt$

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m[u^2 \cos^2 \theta + (u \sin \theta - gt)^2]$$

$$\text{As } v(t) = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$$

$$\text{K.E.} \propto t^2 \text{ parabola}$$

and K.E. $\neq 0$ in projectile

5. A

Sol. KCL at x

$$\frac{x}{R} = \frac{50-x}{R} + \frac{(100-x)-x}{R} = 0$$

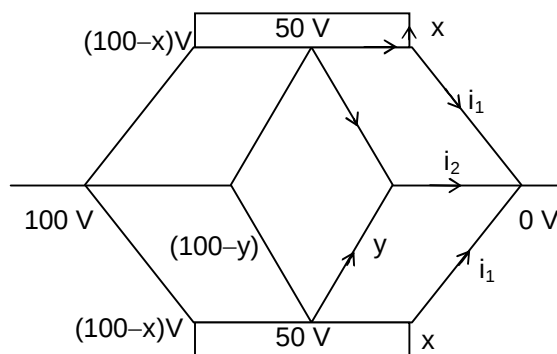
$$\therefore x = \frac{75}{2} \text{ V}$$

and KCL at y

$$\frac{y}{R} = \frac{50-y}{R} + \frac{50-y}{R}$$

$$\therefore y = \frac{100}{3} \text{ V}$$

$$\text{Current} = 2i_1 + i_2 = \frac{2(x)}{R} + \frac{y}{R} = \frac{325}{3R} = \frac{5}{3} \text{ amp}$$



6. D

$$\text{Sol. } \frac{d\theta}{dt} \text{in} = \frac{d\theta}{dt} \text{out}$$

$$\frac{k_1 A_1 (T_1 - T_0)}{2b} = \frac{k_2 A_2 (T_0 - T_2)}{b}$$

$$T_0 = \frac{T_1 + T_2}{2} \text{ and } A_1 = A_2$$

$$\text{We get } k_1 = 2k_2$$

7. D

$$\text{Sol. } I_1 = \frac{I_0}{2}, I_2 = \frac{I_0}{2} \cos^2 30^\circ, I_3 = I_2 \cos^2 30^\circ = \frac{I_0}{2} \cos^4 30^\circ$$

$$I_4 = I_3 \cos^2 30^\circ = \frac{I_2}{2} \cos^6 30^\circ = \frac{I_0}{2} \left(\frac{\sqrt{3}}{2} \right)^6 = \frac{27}{128} I_0$$

8. A

$$\text{Sol. } \frac{V}{R_2} e^{-t/R_2 C} + \frac{V}{R_1} (1 - e^{-R_1 t/L}) = \text{constant}$$

$$\Rightarrow \frac{V}{R_2} e^{-t/R_2 C} = \frac{V}{R_1} e^{-R_1 t/L} \Rightarrow R_1 = R_2$$

$$\frac{R_1}{L} = \frac{1}{R_2 C}$$

$$\Rightarrow R_1 R_2 = \frac{L}{C}$$

9. C

Sol. As the tank is sufficiently deep, the ball will attain terminal velocity

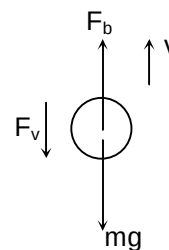
$$v_T = \frac{2(\rho_\ell - \rho_b)R^2g}{9\eta}$$

 ρ_ℓ = density liquid ρ_b = density body

$$v_T \propto (\rho_\ell - \rho_b)R^2$$

$$\therefore v_{T(2)} > v_{T(1)}$$

$$\therefore H_2 > H_1$$



10. D

$$\text{Sol. } d\theta = 1.22 \frac{\lambda}{d}$$

$$d\theta_I = 2.56 \times 10^{-3}$$

$$d\theta_{II} = 3.05 \times 10^{-7}$$

$$d\theta_{III} = 1.22 \times 10^{-7}$$

$$d\theta_{IV} = 6.1 \times 10^{-3}$$

$$\text{Resolving power} = \frac{1}{d\theta}$$

$$RP_{III} > RP_{II} > RP_I > RP_{IV}$$

11. A

Sol. For isothermal Bulk modulus $\propto$ Pressure

12. B

Sol. Using Principle of calorimetry

$$mc \Delta T = 0.01 \text{ mL}$$

$$\text{hence } \Delta T = 0.01 \text{ L/c} = 0.01 \times 2.26 \times 10^6 / 4200 = 5.4^\circ\text{C F.}$$

New temperature (assuming no other losses) is 4.6°C .

13. D

$$\text{Sol. } B_1 + B_2 = B_3$$

$$\frac{\mu_0 I}{2\pi(d+x)} + \frac{\mu_0 I}{2\pi x} = \frac{\mu_0 I}{2\pi(d-x)}$$

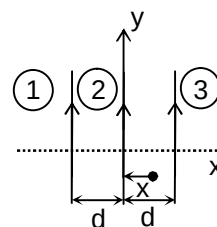
$$\frac{1}{d+x} + \frac{1}{x} = \frac{1}{d-x} \Rightarrow \frac{x+d+x}{x(d+x)} = \frac{1}{(d-x)}$$

$$\Rightarrow (d+2x)(d-x) = dx + x^2$$

$$\Rightarrow d^2 - xd + 2xd - 2x^2 = dx + x^2$$

$$\Rightarrow d^2 + xd - 2x^2 = dx + x^2$$

$$\Rightarrow 3x^2 = d^2 \Rightarrow x = \pm \frac{d}{\sqrt{3}}$$



14. C

$$\text{Sol. } v(x) = v_0 \sin\left(\frac{\pi x}{x_0}\right)$$

$$\Rightarrow a(x) = v_0 \cos\left(\frac{\pi x}{x_0}\right) \frac{\pi}{x_0} \frac{dx}{dt} = v_0 \cos\left(\frac{\pi x}{x_0}\right) \frac{\pi}{x_0} v_0 \sin\left(\frac{\pi x}{x_0}\right)$$

$$a(x) = \frac{\pi v_0^2}{2x_0} \sin\left(\frac{2\pi x}{x_0}\right) \Rightarrow a(x) = \frac{\pi v_0^2}{2x_0} \left(\frac{2\pi x}{x_0}\right) = \frac{\pi^2 v_0^2}{x_0^2} x$$

So, motion is not oscillatory

$$\frac{d^2x}{dt^2} - k^2x = 0 \Rightarrow x = Ae^{\lambda t} + Be^{-\lambda t}, \text{ where } k = \frac{\pi v_0}{x_0}$$

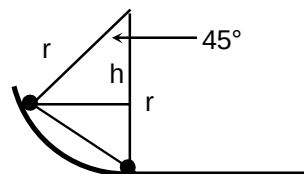
As $t \rightarrow \infty$, $x = Ae^{\lambda t}$ if we assume k small and t is large, we can assume x is some fixed quantity.

15. D

Sol. Electrostatic potential energy $U \propto \frac{1}{r^3} \Rightarrow \frac{U_2}{U_1} = \frac{r_1^3}{r_2^3} = \frac{1}{8}$

16. B

Sol. $h = r - r \cos \theta$
 $= r(1 - \cos \theta)$
 $= 3.4(1 - \cos 45^\circ)$
 $= 1.7(2 - \sqrt{2}) \text{ m}$



and from energy conservation, $mgh = \frac{1}{2} \times 2m \times v^2$

Hence, $v = \sqrt{gh} = \sqrt{9.81 \times 1.7(2 - \sqrt{2})} = 3.1 \text{ ms}^{-1}$

17. D

Sol. $\omega = \sqrt{\frac{k}{m+m'}} = \sqrt{\frac{640}{0.38+0.02}} = \sqrt{1600}$

$\omega = 40 \text{ rad/s}$

Let v' be the velocity acquired by the block m where bullet m strikes it and comes to rest in it. By conservation of momentum

$$(m+m')v' = m'v \Rightarrow v' = \frac{m'}{m+m'}v = \frac{0.02}{0.38+0.02} \times 200 = \frac{0.02}{0.4} \times 200 = 10 \text{ m/s}$$

The block is set in oscillation about its mean position with maximum amplitude A

$$\therefore x = -A \sin \omega t \Rightarrow \frac{dx}{dt} = -A\omega \cos \omega t$$

In the mean position, the velocity is maximum

$$\therefore A\omega = 10 \Rightarrow A = \frac{10}{\omega} = \frac{10}{40} = 0.25$$

$$\therefore x = -0.25 \sin(40t)$$

18. A

Sol. The electric fields at $r < b$ and $r > a$ are both zero. At $b < r < a$ the electric field is

$$E = \frac{Q}{4\pi\epsilon_0 r^2} e_r$$

Hence the field energy is

$$W = \int_b^a \frac{1}{2} \epsilon_0 \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 4\pi r^2 dr = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$

When the outer spherical surface contracts from $r = a$ to $r = a'$, the work done by the electric force is equal to the decrease of the electric field energy

$$W_a - W_{a'} = \frac{Q^2}{8\pi\epsilon_0} \left(-\frac{1}{a} + \frac{1}{a'} \right) = \frac{Q^2(a - a')}{8\pi\epsilon_0 aa'}$$

19. C

Sol. C is the point of contact around which the masses M and m rotate. As it is the instantaneous center of zero velocity, the equation of motion is of the form $\Sigma \tau_C = I_C \alpha$, where I_C is the moment of inertia of masses M and m with respect to point C. Now

$$I_C = \left(\frac{1}{2} MR^2 + MR^2 \right) + md^2 \quad \dots(i)$$

$$\text{Where } d^2 = L^2 + R^2 - 2RL \cos \theta \quad \dots(ii)$$

For small oscillation, $\sin \theta \approx \theta$, $\cos \theta \approx 1$ and

$$I_C = \frac{3MR^2}{2} + m(L - R)^2$$

Therefore, the equation of motion become

$$\left[\frac{3MR^2}{2} + m(L - R)^2 \right] \alpha = -mgL \sin \theta = -mgL\theta$$

$$\text{Or } \alpha + \frac{mgL}{\frac{3MR^2}{2} + m(L - R)^2} \theta = 0$$

$$\therefore \omega = \sqrt{\frac{mgL}{\frac{3MR^2}{2} + m(L - R)^2}} \text{ rad/s}$$

20. D

$$\text{Sol. (i) } v_0 = \frac{X_C}{X_C + X_C} v_i \Rightarrow \frac{v_0}{v_i} = \frac{1}{2}, \text{ phase difference } (\delta) \text{ is } 0$$

$$\text{(ii) } v_0 = \frac{X_C}{R + X_C} v_i \Rightarrow \frac{v_0}{v_i} = \frac{1}{1 + \frac{R}{X_C}} = \frac{1}{1 + i\omega CR} = \frac{1}{\sqrt{1 + (\omega CR)^2}} e^{-i\omega CR} \text{ phase difference } (\delta) \text{ is } 0$$

Phase difference (δ) is $0 < \delta < \frac{\pi}{2}$

SECTION - B

21. 00005.00

$$\text{Sol. } \frac{\sin i}{\sin r} = \sqrt{2} \Rightarrow i = 45^\circ$$

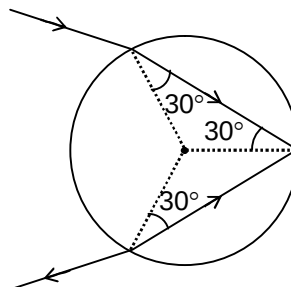
Total deviation

$$= (45^\circ - 30^\circ) + 180^\circ - 2(30^\circ) + (45^\circ - 30^\circ)$$

$$= 30^\circ + 120^\circ$$

$$= 150^\circ = 30x$$

$$\Rightarrow x = 5$$

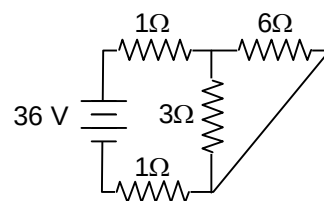


22. 00009.00

Sol. Equivalent circuit can be drawn as following

$$R_{eq} = 4\Omega$$

$$\text{So, } I = 9A$$



23. 00001.50

$$\text{Sol. } \ln V = \ln V_0 - \frac{t}{RC}$$

$$\frac{\Delta V}{V} = 0 - \frac{t}{C} \frac{d}{dR} \left(\frac{1}{R} \right) \Delta R - \frac{t}{R} \frac{d}{dC} \left(\frac{1}{C} \right) \Delta C$$

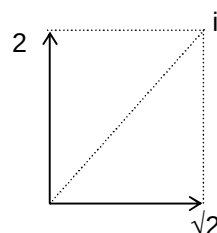
$$= \frac{t}{CR} \left(\frac{\Delta R}{R} + \frac{\Delta C}{C} \right) = \frac{50 \times 10^{-6}}{10^{-8} \times 5 \times 10^4} \left(\frac{5}{100} + \frac{10}{100} \right) = 0.015 = N$$

24. 00001.73

$$\text{Sol. } i = 2 \sin \left(\frac{\pi}{2} + \omega t \right) + \sqrt{2} \sin \omega t$$

$$i = \sqrt{(2)^2 + (\sqrt{2})^2} = \sqrt{6}$$

$$i_{RMS} = \frac{i}{\sqrt{2}} = \frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}$$



25. 00006.00

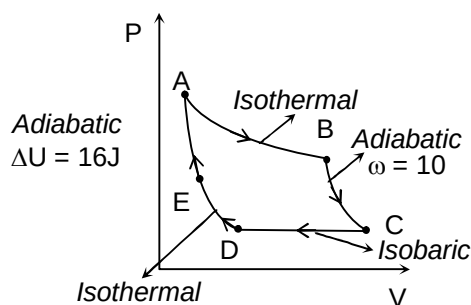
$$\text{Sol. } \Delta U_{AB} = 0, \Delta U_{DE} = 0$$

$$\therefore (\Delta U)_{\text{cycle}} = 0$$

$$\Delta U_{BC} + \Delta U_{CD} + \Delta U_{EA} = 0$$

$$-10 + \Delta U_{CD} + 16 = 0$$

$$\Rightarrow \Delta U_{CD} = -6$$



26. 00002.00

Sol. Time difference (Separation) between two zero velocities

$$\frac{T}{2} = 0.25s, T = 0.5s$$

$$\text{So, frequency } f = \frac{1}{T} = \frac{1}{0.5} = 2 \text{ Hz}$$

27. 00000.50

$$\text{Sol. } \lambda = \frac{1}{\sqrt{2n\pi d^2}} = \frac{V}{\sqrt{2N\pi d^2}}$$

$$\Rightarrow \lambda \propto \frac{V}{N} \Rightarrow V \propto \lambda N$$

$$\frac{V_1}{V_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2} = \frac{\lambda_1 N_1}{(4\lambda_1) \left(\frac{N_1}{2} \right)} = 0.5$$

28. 00014.14

Sol. $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$

$$y = 5 \text{ m}$$

velocity just before collision $\vec{v} = 10\hat{i} + 0\hat{j}$

Apply elastic collision, velocity $\vec{v}_x = 10(-\hat{i}) \text{ m/s}$

$$\vec{v}_y = 0$$

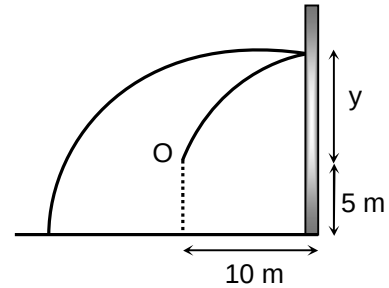
$$s = ut + \frac{1}{2}at^2 \text{ in y-axis}$$

$$\Rightarrow -(y+5) = 0 - \frac{1}{2}(10)t^2$$

$$\Rightarrow -10 = -\frac{10}{2}t^2$$

$$t = \sqrt{2} \text{ sec}$$

Range after collision in x-axis $(v_x)t = 10\sqrt{2} = 14.14 \text{ m}$



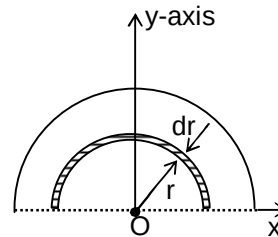
29. 00001.60

Sol. $dm = \rho \pi r dr = c r^2 \pi r dr = c \pi r^3 dr$

$$y_{cm} = \frac{\int dm \frac{2r}{\pi}}{\int dm}$$

$$y_{cm} = \frac{\int_0^R c \pi r^3 dr \frac{2r}{\pi}}{\int_0^R c \pi r^3 dr \frac{2r}{\pi}} = \frac{2 \int_0^R r^4 dr}{2 \int_0^R r^3 dr} = \frac{\frac{2}{5} R^5}{\frac{2}{4} R^4} = \frac{\pi}{5} \frac{R^5}{R^4}$$

$$y_{cm} = \frac{8R}{5\pi} = \frac{8}{5}$$



30. 00004.04

Sol. Let the body travel a distance s on the incline and come down through a height h .

Potential energy lost $= mgh = mgs \sin \theta$

Work done against friction $W = fs = \mu mg \cos \theta \cdot s$

By problem $\mu mg \cos \theta s = \frac{70}{100} mgs \sin \theta$

$\therefore \mu = 0.70 \tan \theta = 0.7 \tan 30^\circ = 0.404$

Chemistry

PART – B

SECTION – A

31. C

Sol. From the chemical equation, we can have

$$\frac{1}{2} \times 0.15 \times 35 \times 10^{-3} \text{ L} = \frac{1}{5} \times [\text{H}_2\text{O}_2] \times 50 \times 10^{-3} \text{ L}$$

$$\therefore [\text{H}_2\text{O}_2] = 0.263 \text{ M}$$

32. D

Sol. The mass% of nitrogen is higher in their compound hence gives greatest amount of N_2 gas.

33. B

Sol. As we know that

$$\Delta G = \Delta H - T\Delta S = -RT \ln K$$

Here the reaction is vapourisation, therefore, the equilibrium constant equal to vapour pressure of the liquid

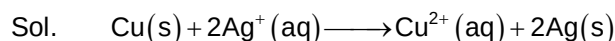
$$\text{Thus, } \ln P = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

Slope of the graph in the question is equal to $-\frac{\Delta H}{R}$

$$\therefore \Delta H = -4544 \times -8.314$$

$$= 37.8 \text{ kJ/mol.}$$

34. D



By Nernst equation

$$E = E^0 - \frac{RT}{nF} \ln Q$$

$$\therefore Q = \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$$

$\therefore$ Increasing Q will decrease E.

Hence, I is incorrect and by adding Cl^- to the Ag^+ .

$\therefore$ Cell precipitate

$\therefore [\text{Ag}^+] \downarrow$ es hence Q is $\uparrow$ es

And $\therefore$ II is also incorrect.

35. C

Sol. $\text{Ge} < \text{Si} < \text{As} < \text{P}$

$\text{Si} = 786 \text{ kJ/mole}$; $\text{P} = 1012 \text{ kJ/mole}$; $\text{Ge} = 761 \text{ kJ/mole}$; $\text{As} = 947 \text{ kJ/mole}$.

36. A

Sol. Isoelectric point is significantly determined by its amine and carboxylic acid functionalities.

37. A
Sol. Si – O bond require more energy becomes it is stronger than Si – Si bond, as 'O' atom is smaller than Si.
Si is not a metal and SiO₂ is not a ionic compound and also not polar but Si and SiO₂ forms 3D network covalent solid.
38. A
Sol. Conc. HNO₃ or HCl cannot oxidize Au to Au³⁺ because standard reduction potential Au³⁺ + 3e⁻ → Au is too high.
But Cl⁻ in aqua regia complex with Au³⁺, lowering down the reduction potential through stabilizing it as AuCl₄⁻.
39. C
Sol. Most basic, hence easily alkylation.
40. A
Sol.
$$\frac{(4 \times 23 + 4x) \times \frac{1}{6.023 \times 10^{23}}}{\left(462 \text{ pm} \times \frac{1 \text{ cm}}{10^{10} \text{ Pm}}\right)^3} = 2.83 \text{ g/cm}^3$$
$$\frac{92 + 4x}{6.023 \times 10^{23}} = 2.83 \times (462 \times 10^{-10})^3$$
$$92 + 4x = 168$$
$$x = \frac{168 - 92}{4} = \frac{76}{4} = 19$$
$$\therefore \text{NaF}$$
41. B
Sol. % hydrolysis does not depend on the concentration, in case of weak acid + weak base; salt.
42. A
Sol. $\text{NaCl} + \text{AgNO}_3 \longrightarrow \text{AgCl} \downarrow + \text{NaNO}_3$
With ppt.
 $\text{AgCl} + \text{NH}_4\text{OH} \longrightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 $[\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{HNO}_3 \longrightarrow \text{AgCl} \downarrow + \text{NH}_4\text{NO}_3$
43. C
Sol. $n_2^2 - n_1^2 = 8$
$$(n_2 - n_1)(n_2 + n_1) = 8$$
$$n_2 - n_1 = \frac{8}{4} = 2$$
$$n_2 + n_1 = 4$$
$$\therefore n_2 = 3$$
$$n_1 = 1$$
$$\frac{1}{\lambda} = R_H \times 4 \left[1 - \frac{1}{9}\right]$$

$$\lambda = \frac{9}{32R_H}$$

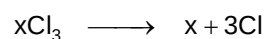
44. A

Sol. $\frac{P_0 - P_s}{P_s} = m \times \text{mol. wt. of solution}$

$$\frac{P_0 - P_s}{P_s} \times \frac{1000}{18} = m$$

$$m = \frac{17.25 \times 17.20}{17.20} \times \frac{1000}{18} = 0.1615$$

If $m = M = 0.1615$.



$$M_{\text{total}} \quad s + 3s = 4s = 0.1615$$

$$s = \frac{0.1615}{4} = 0.04 \times 10^{-2}$$

$$= 4.037 \times 10^{-2}$$

45. C

Sol. Neutral species is more stable than charged and complete octet resonance is more stable than incomplete octet resonance.

46. D

Sol. Sulphuric acid does not show efflorescent.

47. B

Sol. $b = 4V_m$

$$= 4 \times N_A \times \frac{4}{3} \pi r^3$$

$$V_1 = V - nb$$

$$nb = V - V_1$$

$$b = \frac{V - V_1}{n} = 4.N_A \times \frac{4}{3} \pi r^3$$

$$r = \left[\frac{3(V - V_1)}{16\pi N_A n} \right]^{1/3}$$

$$D = 2 \left[\frac{3(V - V_1)}{2 \times 2^3 \pi N_A n} \right]^{1/3}$$

$$D = \left[\frac{3(V - V_1)}{2\pi N_A n} \right]^{1/3}$$

48. B

Sol. The refining process of blister copper is known as polling and Cu_2O impurities are present in the Cu metal.

49. B

Sol. Meq. of complex = Meq. Of acid = Meq. of NaOH

$$\begin{aligned}\frac{W}{E} \times 1000 &= N \times V \\ &= 0.1 \times 20 \\ \frac{W}{M} \times nf &= \frac{0.1 \times 20}{1000} \\ n \times nf &= \frac{0.1 \times 20}{1000} \\ nf &= \frac{0.1 \times 20}{0.001 \times 1000} = 2\end{aligned}$$

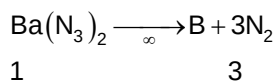
50.

B

Sol.

$$PV = nRT$$

$$n = \frac{RV}{RT} = \frac{1.23 \times 20}{0.0821 \times 300} = 1$$



Number of moles $\text{Ba}(\text{N}_3)_2$ required for 1 mole of $\text{N}_2 = \frac{1}{3}$ mole

$$= \frac{1}{3} \times (137.327 + 84)$$

$$= \frac{221.327}{3} \approx 74$$

SECTION – B

51.

00080.10

Sol.

By law of conservation of environment

$$\theta_t = \theta_1 + \theta_2$$

$$-200 = 10(0.461)(T - 100) + 10(0.544)(T - 100)$$

$$= (T - 100)(0.461 + 0.544) \times 10$$

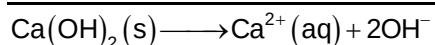
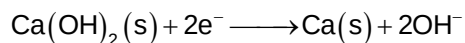
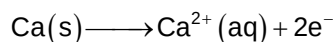
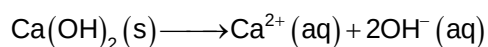
$$T = 80.1^\circ\text{C}$$

52.

00008.44

Sol.

The dissociation reaction of $\text{Ca}(\text{OH})_2$ is



$$E^0 = E_{\text{cal}}^0 - E_{\text{anode}}^0$$

$$= -3.02 \text{ V} - (-2.87 \text{ V}) = -0.15$$

From Nernst equation

$$E = E^0 - \frac{RT}{nF} \ln Q$$

At equilibrium $E = 0$

$$E^{\circ} = \frac{RT}{nF} \ln K_{sp}$$

$$\ln K_{sp} = \frac{E^{\circ} \cdot nF}{RT}$$

$$\log K_{sp} = \frac{E^{\circ} \cdot nF}{2.303RT} = \frac{-0.15 \times 2 \times 96500}{2.303 \times 8.314 \times 298}$$

$$\log K_{sp} = \frac{-28950}{5705.848} = -5.0737$$

$$\log K_{sp} = -5.0737$$

$$K_{sp} = \text{Anti log}(-5.0737)$$

$$= \text{Antilog } \bar{6}.9263$$

$$K_{sp} = 8.44 \times 10^{-6}$$

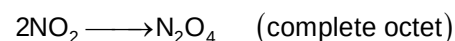
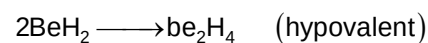
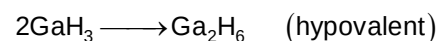
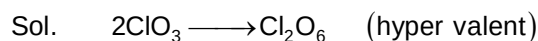
53. 00001.86

Sol. $K_P = \frac{[P_{SO_2}]^1}{[P_{SO_2}][PO_2]^{1/2}} = \frac{0.332}{(0.562)(0.101)^{1/2}} = 1.86$

54. 00005.97

Sol. $PI = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{2.34 + 9.60}{2} = \frac{11.94}{2} = 5.97$

55. 00003.00

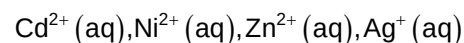


$$\therefore x = 2 \mid y = 3 \mid z = 2$$

$$\therefore \frac{yz}{x} = \frac{3 \times 2}{2} = 3$$

56. 00004.00

Sol. $\therefore$ Soluble in excess of NH_3 sol.



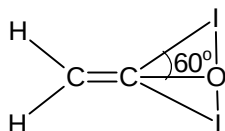
57. 00005.00

Sol. Number of electron in non-axial d-orbital

$$(d_{xy}, d_{yz}, d_{xz}) = 2 + 2 + 1 = 5$$

58. 00004.07

Sol.



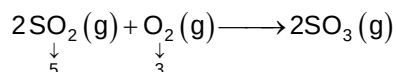
$$IO = C - I \sin 60^\circ = 4.07$$

$$\therefore I-I = 2 \times IO$$

$$= 2 \times 2.035 = 4.07$$

59. 00005.50

Sol.



Limiting reagent is SO_2 , hence total volume of gas after complete formation of $\text{SO}_3 = 5 \text{ L } \text{SO}_3 + 0.5 \text{ L } \text{O}_2 = 5.5 \text{ L}$.

60. 00007.00

Sol.

$$K = (n-5)s^2$$

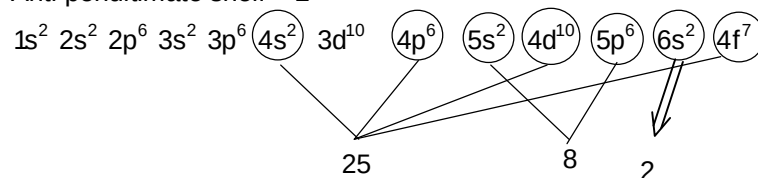
 $\downarrow$

6

Outer most shell = 2

Penultimate shell = 8

Anti penultimate shell = 2



Mathematics

PART – C

SECTION – A

61. B

Sol. Consider the three-digit arrangement, aba. There are 10 choices for a and 10 choices for b (since it is possible for a = b), and so the probability of picking the palindrome is $\frac{10 \times 10}{10^3} = \frac{1}{10}$

Similarly, there is a $\frac{1}{26}$ probability of picking the three-letter palindrome

By the Principle of inclusion-exclusion, the total probability is $\frac{1}{26} + \frac{1}{10} - \frac{1}{260} = \frac{35}{260} = \frac{7}{52}$
 $\Rightarrow 7 + 52$

62. C

Sol. By property of inclusion/exclusion, we have $|A_1 \cup A_2 \cup A_3| = \sum |A_i| - \sum |A_i \cap A_j| + |A|$

Number of people in at least two sets is $\sum |A_i \cap A_j| - 2|A_1 \cap A_2 \cap A_3| = 9$

So, $20 = (10 + 13 + 9) - (9 - 2x) + x$, which gives x

63. B

Sol. Let G be the point obtained by reflecting C about the line AB. Since $\angle ADC = \frac{\pi - \theta}{2}$,

we find that $\angle BDE = \pi - \theta - \angle ADC = \frac{\pi - \theta}{2} = \angle ADC = \pi - \angle BDC = \pi - \angle BDC$

$|EF| = \frac{|BE|}{|BC|} = \frac{|BE|}{|BG|} = \frac{\sin\left(\frac{\theta}{2}\right)}{\sin\left(\frac{3\theta}{2}\right)}$ where we have used the law of sines in $\triangle BDG$. But by L'Hospital's

Rule

$$\lim_{\theta \rightarrow 0} \frac{\sin\left(\frac{\theta}{2}\right)}{\sin\left(\frac{3\theta}{2}\right)} = \lim_{\theta \rightarrow 0} \frac{\cos\left(\frac{\theta}{2}\right)}{\cos\left(\frac{3\theta}{2}\right)} = \frac{1}{3}$$

64. A

Sol. Since $\sin 2x = 2 \sin x \cos x$ and $\sin 3x = \sin x (4 \cos^2 x - 1)$, we may write the equation in the form $\sin x (2 \cos^2 x - (2 - a) \cos x - a) = 0$. Thus $\sin x = 0$, $\cos x = 1$ or $\cos x = -\frac{a}{2}$. The nonnegative solutions of the equations $\sin x = 0$ and $\cos x = 1$ are $x = k\pi$ and $x = 2k\pi$, $k = 0, 1, 2, \dots$ respectively. Let $|a| > 2$. In this case the equation $\cos x = -\frac{a}{2}$ has no solution and therefore the nonnegative solutions of the initial equation are $0, \pi, 2\pi, \dots$, which form an arithmetic progression. Let now $|a| \leq 2$ and let x_0 be the only solution of the equation $\cos x = -\frac{a}{2}$ in the interval $[0, \pi]$. In this case the nonnegative solutions of the last equation are $x = x_0 + 2k\pi$ and $x = 2\pi - x_0 + 2k\pi$. It is clear now that the nonnegative solutions form an arithmetic progression only when $x_0 = 0$, $x_0 = \frac{\pi}{2}$ and $x_0 = \pi$, so giving $a = -2$, $a = 0$ and $a = 2$. The values of a are $a = 0$ and $|a| > 2$.

65. D

Sol. $z = e^{i\pi/13} \Rightarrow z^{13} = -1 ; z^{13} + 1 = 0 \Rightarrow \frac{1+z^{13}}{1+z} = 0$

$$\Rightarrow 1 - z + z^2 - z^3 + \dots + z^{12} = 0 \quad \dots (1)$$

$$\Rightarrow \frac{1-z^{13}}{1-z} = \frac{2}{1-z}$$

$$\Rightarrow 1 + z + z^2 + z^3 + \dots + z^{12} = \frac{2}{1-z} \quad \dots (2)$$

From equation (1) and (2), we get

$$1 + z^2 + z^4 + z^6 + z^8 + z^{12} = \frac{1}{1-z} \text{ and } z + z^3 + z^5 + \dots + z^{11} = \frac{1}{1-z}$$

66. D

Sol. $|2z_1 - 3z_2 - 4z_3| = |2z_1 - 3(z_2 - 2) - 4(z_3 - 5) - 26| \leq 2|z_1| + 3|z_2 - 2| + 4|z_3 - 5| + 26 = 61$

67. A

Sol. $\lim_{x \rightarrow 0} \left(3 - \frac{P(x)}{x} \right) = 27 \Rightarrow P(x) \text{ has no constant term let } P(x) = ax^4 + bx^3 + cx^2 + dx$

$$\Rightarrow 3 - d = 27 \Rightarrow d = -24$$

$$P(x) = ax^4 + bx^3 + cx^2 - 24x ; P'(2) = 0, p(1) = -9, p'''(2) = 0$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx - 24 ; P''(x) = 12ax^2 + 6bx + 2c$$

$$P'''(x) = 24ax + 6b ; a + b + c - 24 = -9$$

$$\Rightarrow a + b + c = 15 \quad \dots (1)$$

$$P'(2) = 0$$

$$\Rightarrow 4a(8) + 3b(4) + 2c(2) - 24 = 0$$

$$\Rightarrow 8a + 3b + c = 6 \quad \dots (2)$$

$$P'''(2) = 0$$

$$\Rightarrow 24a(2) + 6(b) = 0 \Rightarrow 8a + b = 0 \quad \dots (3)$$

Solving (1), (2) and (3)

$$a = 1, b = -8, c = 22$$

$$\Rightarrow P(x) = x^4 - 8x^3 + 22x^2 - 24x ; P'(x) = 4x^3 - 24x^2 + 44x - 24 = 4[x^3 - 6x^2 + 11x - 6]$$

$$P'(x) = 4[(x-1)(x-2)(x-3)]$$

$$P''(x) = 4[3x^2 - 12x + 11] > 0 \forall x \in [3, 4] = 4[(3)(2)(1)] = 24 = 4M \Rightarrow M = 6$$

68. A

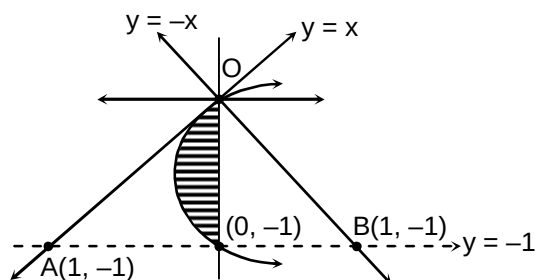
Sol. Tangent to parabola at origin is $x - y = 0$

$$y^2 + y = x ; \left(y + \frac{1}{2} \right)^2 = x + \frac{1}{4}$$

Let A be the shaded area

$$A = \int_{-1}^0 -(y^2 + y) dx = - \left[\frac{y^3}{3} + \frac{y^2}{2} \right]_{-1}^0$$

$$= - \left[- \left(-\frac{1}{3} + \frac{1}{2} \right) \right] = \frac{1}{6}$$



$$\text{Area of OAB} = \frac{1}{2} \times 1 \times 2 = 1 ; \frac{A_1}{A_2} = \frac{\frac{1}{2} - \frac{1}{6}}{\frac{1}{2} + \frac{1}{6}} \text{ or } \frac{\frac{1}{2} + \frac{1}{6}}{\frac{1}{2} - \frac{1}{6}} \Rightarrow \frac{A_1}{A_2} = \frac{1}{2} \text{ or } \frac{2}{1} \Rightarrow \frac{p}{q} = \frac{1}{2} \text{ or } \frac{2}{1} \Rightarrow p + q = 3$$

69. B

Sol. Equation of AB $y = r : A(R - \sqrt{R^2 - r^2}, r)$ using $R^2 = 2r^2$
 $B(R + \sqrt{R^2 - r^2}, r)$ point $A(R - r, r)$ $B(R + r, r)$ slope $OA = \frac{r}{R - r} = m_1$
 Slope $OB = \frac{r}{R + r} = m_2$; $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = 1$; $\theta = \frac{\pi}{4} = 45^\circ$

70. A

Sol. $\int_0^\infty e^{-1}|x-t|dt = \int_0^x e^{-1} \cdot (-t+x)dt + \int_x^\infty e^{-1} \cdot (-x+t)dt$ apply integration by parts
 $= [(x-t)(-e^{-1}) - (-1)e^{-t}]_0^x + [(x-t)e^{-t} - e^{-t}]_x^\infty = x + 2e^{-x} - 1$

71. C

Sol. $-(1^3 + 2^3 + 3^3 + \dots + 2014^3) + 2(2^3 + 4^3 + \dots + 2014^3)$
 $= -\left(\frac{2014 \cdot 2015}{2}\right)^2 + 2 \cdot 2^3 \left(\frac{1007 \cdot 1008}{2}\right)^2 = -(1007)^2 (2015)^2 + 4(1007)^2 (1008)^2$
 $= -(1007)^2 [(2016)^2 - (2015)^2] = (1007)^2 4031$

72. C

Sol. $f(x) = x^3 + 3x^2 + 3x + 3 = 0$; $f'(x) = 3x^2 + 6x + 3 = 3(x+1)^2 \geq 0$; $f(x)$ is increasing function
 Now $f(-3) = -6 < 0$; $f(-2) = 1 > 0$
 $\therefore$ real root γ lies between -3 and -2
 $\alpha + \beta + \gamma = -3$; $-3 < \gamma < -2$; $\alpha + \beta - 3 < \alpha + \beta + \gamma < \alpha + \beta - 2$
 $\alpha + \beta - 3 < -3 < \alpha + \beta - 2 - 1 < \alpha + \beta < 0$

73. B

Sol. $\frac{a}{4} = \frac{9}{b} \Rightarrow ab = 36$ using $AM \geq GM$
 $\frac{a+b}{2} \geq \sqrt{ab} \Rightarrow a + b \geq 12$

74. A

Sol. Let r be an integer in $(-10, 10)$

$$\begin{aligned} \text{Now, LHL} &= \lim_{x \rightarrow r^-} \int_{-10}^x 2^{[t]} dx = \lim_{h \rightarrow 0^+} \left[\int_{-10}^{-9} 2^{[t]} dt + \int_{-9}^{-8} 2^{[t]} dt + \dots + \int_{r-1}^{r-h} 2^{[t]} dt \right] \\ &= \lim_{h \rightarrow 0^+} [2^{-10} + 2^{-9} + \dots + 2^{r-1}(1-h)] \\ &= 2^{-10} + 2^{-9} + \dots + 2^{r-1} \quad \dots (1) \\ \lim_{x \rightarrow r^+} \int_{-10}^x 2^{[t]} dt &= \lim_{h \rightarrow 0^+} \left[\int_{-10}^{-9} 2^{[t]} dt + \int_{-9}^{-8} 2^{[t]} dt + \dots + \int_r^{r+h} 2^{[t]} dt \right] \\ &= 2^{-10} + 2^{-9} + \dots + 2^{r-1} \quad \dots (2) \\ f(r) &= \int_{-10}^r 2^{[t]} dt = 2^{-10} + 2^{-9} + \dots + 2^{r-1} \quad \dots (3) \end{aligned}$$

From equation (1), (2) and (3), we get $f(x)$ is continuous at all integers

75. A

Sol.
$$\left[(1-x)^2 + 3x\right]^n = \sum_{r=0}^n a_r x^r \left((1-x)^2\right)^{n-r}$$

$$\Rightarrow \sum_{r=0}^n {}^nC_r \left((1-x)^2\right)^{n-r} (3x)^r = \sum_{r=0}^n a_r x^r \left((1-x)^2\right)^{n-r}$$

Compare the coefficients of $x^r (1-x)^{2n-2r}$

76. B

Sol.
$$\sin^{-1}(\sin x) = \begin{cases} x & ; -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ \pi - x & ; \frac{\pi}{2} \leq x \leq \frac{3\pi}{2} \end{cases}$$

77. D

Sol. $R.N.W = {}^7C_2 \times ({}^5C_2 \times 2^2) = 840$

78. A

Sol. $-4x dx + 6y dy + 5(x dy + y dx) - 3dx - 2dy = 0$
Integrating $-2x^2 + 3y^2 + 5xy - 3x - 2y = c$

79. B

Sol. $p \Rightarrow q$
Contrapositive $\sim p \Rightarrow \sim q$

80. A

Sol. $l^2 + m^2 + n^2 = 1 ; l + m + n = 0 ; 2l^2 + 2m^2 - n^2 = 0$

Solving above equations, we get the direction cosines as $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}}\right)$ and $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}}\right)$

or $\left(-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}}\right)$ and $\left(-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}}\right)$

The angle between these lines in both the cases is $\cos^{-1}\left(-\frac{1}{3}\right)$

SECTION – B

81. 00012.00

Sol. Let $y = x \sin x$. We can rewrite the expression as $\frac{9y^2 + 4}{y} = 9y + \frac{4}{y}$

Since, $x > 0$ and $\sin x > 0$ because $0 < x < \pi$, we have $y > 0$. So we can apply AM – GM

$$9y + \frac{4}{y} \geq 2\sqrt{9y \cdot \frac{4}{y}} = 12. \text{ The equality holds when } 9y = \frac{4}{y} \iff y^2 = \frac{4}{9} \iff y = \frac{2}{3}$$

Therefore, the minimum value is 012.

82. 00089.00

Sol. Use the angle bisector theorem to the angle bisector of $\angle P$ divides QR into segments of length

$$\frac{25}{x} = \frac{15}{20-x} \Rightarrow x = \frac{25}{2}, \frac{15}{2}. \text{ It follows that } \frac{QP'}{RP'} = \frac{5}{3} \text{ and so}$$

$$P' = \left(\frac{5x_R + 3x_Q}{8}, \frac{5y_R + 3y_Q}{8} \right) = \left(-5, -\frac{23}{2} \right)$$

The desired answer is the equation of the line PP' . PP' has slope $-\frac{11}{2}$, from which the equation to be $11x + 2y + 78 = 0$. Therefore, $a + c = 089$

83. 00994.00

$$\text{Sol. } a^2 + b^2 - 2ab \cos \gamma = c^2 = 1989c^2 - 2ab \cos \gamma \Rightarrow ab \cos \gamma = 994c^2$$

$$\text{Now, } \frac{\cot \gamma}{\cot \alpha + \cot \beta} = \frac{\cos \gamma \sin \alpha \sin \beta}{\sin \gamma} = \frac{\cos \gamma \sin \alpha \sin \beta}{\sin^2 \gamma} = \frac{ab}{c^2} \cos \gamma = \frac{ab}{c^2} \cdot \frac{99c^2}{ab} = 994$$

84. 00015.00

Sol. It is best to get rid of the absolute values first. Under the given circumstance, we notice that

$$|x - p| = x - p, |x - 15| = 15 - x, \text{ and } |x - p - 15| = 15 + p - x$$

Adding these together, we find that the sum is equal to $2(30 - x)$, which attains its minimum value (on the given interval $p \leq x \leq 15$) when $x = 15$, giving a minimum of 015

85. 00003.00

$$\text{Sol. } A^{-1} + B^{-1} = (A + B)^{-1} \Rightarrow (A^{-1} + B^{-1})(A + B) = (A + B)^{-1}(A + B)$$

$$\Rightarrow 2I + A^{-1}B + B^{-1}A = I \Rightarrow A^{-1}B + B^{-1}A = -I$$

$$\text{Let } A^{-1}B = P$$

$$P + P^{-1} = -I \Rightarrow P^2 + I + -P \Rightarrow P^2 + I = -P \Rightarrow P^2 + P + I = 0$$

$$\Rightarrow (P - I)(P^2 + P + I) = 0 \Rightarrow P^3 = I$$

$$|B| = 1 \Rightarrow |A^{-1}| |B| = 1 \Rightarrow |A| = |B|$$

86. 00016.00

$$\text{Sol. } A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a^2 + bc & (a+d)b \\ c(a+d) & bc + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{If } a + d = 0, a^2 + bc = 1, b = 0, c = 0, d = 1, -1, a = 1, -1$$

$$\text{or } c = 0, b = 1, -1, a = 1, -1$$

$$\text{If } b = 0, c = 0, a = 1, -1, d = 1, -1$$

$$\text{Number of quadruples } (a, b, c, d) = 16$$

87. 00016.43

$$\text{Sol. } \frac{2}{\sqrt{30}}(3+3) + \frac{5}{\sqrt{30}}(8+7) + \frac{-1}{\sqrt{30}}(3-6) = 3\sqrt{30}$$

88. 00001.00

$$\text{Sol. } m^3 = -\frac{4}{49} = -\frac{1}{9}$$

89. 00120.00

$$\text{Sol. } x + y + z \leq 10 \Rightarrow \text{R.N.W} = {}^9C_2 + {}^8C_2 + \dots + {}^2C_2 = 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120$$

90. 00005.00

Sol. Let the variable be x . Here $n = 100$, incorrect $A = 40$, incorrect $\sigma = 5.1$

From formula $A = \frac{\sum x}{n} \cdot \sum x = nA$

$\therefore$ Incorrect $\sum x = 100 \times 40 = 4000$

$\therefore$ Correct $\sum x = \text{incorrect } \sum x - 50 + 40 = 4000 - 50 + 40 = 3990$

Again, from $\sigma^2 + A^2 = \frac{\sum x^2}{n}$, we get $\sum x^2 = n(\sigma^2 + A^2)$

$\therefore$ Incorrect $\sum x^2 = 100\{(5.1)^2 + (40)^2\} = 100(26.01 + 1600) = 162601$

$\therefore$ Correct $\sum x^2 = \text{incorrect } \sum x^2 - 50^2 + 40^2 = 162601 - 2500 + 1600 = 161701$

$\therefore$ Correct mean $A = \frac{\sum x}{n} = \frac{3990}{100} = 39.9$

Correct $(\sigma^2 + A^2) = \frac{\sum x^2}{n} = \frac{161701}{100} = 1617.01$

$\therefore$ Correct $\sigma^2 = 1617.01 - (\text{correct } A^2) = 1617.01 - (39.9)^2 = 1617.01 - 1592.01 = 25$